Note 1.6 Inverse Functions and Inverse Trigonometric Functions

One-to-One Functions

- **Definition:**
  A function $f(x)$ is one-to-one on a domain D if and only if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D.
- **The Horizontal Line Test for One-to-One Functions**
  A function $y = f(x)$ is one-to-one if and only if its graph intersects each horizontal line at most once.
  Note:
  increasing/decreasing $\implies$ one-to-one
  one-to-one $\neq>$ increasing/decreasing

Inverse Functions

- **Definition:**
  If the function f is one-to-one on a domain D with range R, then the inverse function f^{-1} is defined by

$$f^{-1}(b) = a \text{ if } f(a) = b$$

  The domain of f^{-1} is R and the range of f^{-1} is D.
- **Remarks:**
  - f is increasing/decreasing $\implies$ f^{-1} increasing/decreasing
  - f and f^{-1} is symmetric about $y = x$.

Inverse Trigonometric Functions

- Restrict their domains to intervals on which they are one-to-one.
- Domain restrictions that make the trigonometric functions one-to-one

The Arcsine and Arccosine Functions

- **Definition:**
  $y = \sin^{-1} x$ is the number in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for which $\sin y = x$
  $y = \cos^{-1} x$ is the number in $[0, \pi]$ for which $\cos y = x$
- **Identity:**
  - $\sin^{-1}(-x) = -\sin^{-1} x$
  - $\cos^{-1} x = \pi - \cos^{-1}(-x)$

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

Elementary Functions

- Constant functions, power functions, exponential functions, logarithmic functions, trigonometric functions, inverse trigonometric functions and the functions combining them. (addition, subtraction, multiplication, quotient, composition)